

Department of Mathematics and Computing
National Institute of Technology Kurukshetra

Total Marks: 50

MAIC 207: Optimization Techniques (End Sem)
Use of Calculators is allowed, Time: 2:00 PM to 05:00 PM, Date: 09/08/2026

- ✓ 1. (4M) A factory is engaged in manufacturing two products A and B which involve lathe work, grinding and assembling. The cutting, grinding and assembling times required for one unit of A are 2, 1 and 1 hours respectively and for one unit of B are 3, 1 and 3 hours respectively. The profits on each unit of A and B are Rs. 2.00 and Rs. 3.00 respectively. Assuming that 300 hours of lathe time, 300 hours of grinding time and 240 hours of assembling time are available, pose a linear programming problem in terms of maximizing the profit on the items manufactured.

- ✓ 2. (8M) Use Simplex method to solve the following LPP.

$$\begin{aligned} \text{Maximize } z &= 2x_1 + 3x_2 + x_3 \\ \text{subject to } -3x_1 + 2x_2 + 3x_3 &= 8 \\ -3x_1 + 4x_2 + 2x_3 &= 7 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

3. How does Simplex method detect:

(i) (1.5M) Alternate optima

unbounded
optimal

(ii) (1.5M) No solution.

4. (1M+4M) Define convex function. Let $f, g: \mathbb{R}^n \rightarrow \mathbb{R}$ be two convex functions. Prove/disprove that $f + g$ and $f - g$ are convex.

- ✓ 5. (4M) Find the critical points and classify them as local maxima, local minima, and saddle point of the following function.

$$f(x, y) = 2xy - 8x - 2y^2 + 3x + 4.$$

6. Answer the following questions.

- ✓ (i) (4M) Suppose the loss function for a machine learning model is

$$L(w) = w^2 - 6w + 10.$$

Starting with $w_0 = 5$, perform two iterations of the steepest descent algorithm with learning rate $\alpha = 0.2$.

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